

NORMENS[®] 5mg Tablet

(Norethisterone acetate)

Presentation:

Each white, uncoated tablet contains Norethisterone acetate BP equivalent to 5mg norethisterone.

Product Description:

6.0 mm, White, Round Flat Bevel Edge tablet, one side having logo “N” and other side contains breakline.

Pharmacokinetics

• Absorption

Peak serum concentrations after administration of 5mg **NORMENS[®]** were 16 ± 5 ng/mL; $t_{\max}$ was 1.5 ± 0.6 hours. The bioavailability of a single dose of 1mg norethisterone compared to an equivalent intravenous dose ranged from 47% to 73%, suggesting a significant first pass effect. This study showed that the disposition of norethisterone follows a biexponential pattern with half lives of 0.4-2.6 and 5-13 hours respectively. The apparent volume of disposition was 4.28 ± 0.56 L/kg.

• Distribution

Norethisterone is bound to serum albumin and to sex hormone binding globulin (SHBG). Only about 3-4 % of the total serum drug concentrations are present as free steroid, about 35 % and 61 % are bound to SHBG and albumin, respectively. The apparent volume of distribution of norethisterone is 4.4 ± 1.3 L/kg. Following oral administration, the drug serum level time course follows a biphasic pattern. Both phases are characterized by half-lives of 1-2 and about 5-13 hours, respectively. Norethisterone is transferred into milk and the drug levels in breast milk were found to be about 10% of those found in maternal plasma.

• Metabolism

Norethisterone is mainly metabolized by saturation of the double bond in ring A and the reduction of the 3-keto group to a hydroxyl group followed by conjugation to the corresponding sulfates and glucuronides. Some of these metabolites are eliminated slowly from plasma, building approximately to a peak at the midpoint of the treatment cycle and rapidly dropping to near baseline levels when treatment is stopped. Norethisterone is partly metabolized to ethinylestradiol. Per one milligram of orally administered norethisterone, ethinylestradiol is formed equivalent to an oral dose of approximately 4 µg in humans. Since

the estrogenicity of norethisterone has always been assumed and experienced in clinical practice, the recent discovery of its metabolic characteristics does not change the existing recommendations for use.

- **Elimination**

Norethisterone is not excreted unchanged to a significant extent. Predominantly A-ring-reduced and hydroxylated metabolites as well as their conjugates (glucuronides and sulfates) are excreted via urine and feces. 37.4% to 80% of the orally administered dose was excreted in the urine over 5 days with a half-life of about 19 hours.

- **Steady-state conditions**

During multiple-dose daily administration with norethisterone, an accumulation of the drug is unlikely because of the relatively short half-life of the drug. If, however, SHBG inducing agents such as ethinylestradiol are co-administered, an increase in norethisterone serum levels can occur because of the binding of norethisterone to SHBG.

Pharmacodynamics:

Norethisterone is a progestogen with negligible androgenic effects. Complete transformation of the endometrium from a proliferative to a secretory state was achieved in estrogenprimed castrated or climacteric women who were administered oral doses of 100-150 mg norethisterone per cycle. The progestogenic effects of norethisterone on the endometrium is the basis of the treatment of dysfunctional bleeding, primary and secondary amenorrhea, and endometriosis with **NORMENS®**. Gonadotropin secretion inhibition and anovulation can be achieved with a daily intake of 0.5 mg of norethisterone. Positive effects of Norethisterone on premenstrual symptoms can be traced back to suppression of ovarian function.

Due to the stabilizing effects of norethisterone on the endometrium, administration of Norethisterone can be used to shift the timing of menstruation. Like progesterone, norethisterone is thermogenic and alters the basal body temperature.

Indications

Dysfunctional Uterine Bleeding (DUB), Premenstrual syndrome, Postponement of menstruation, Endometriosis, Prevention of lactation, Mastodynia.

Pregnancy and Lactation:

The use of norethisterone during pregnancy is contraindicated. Norethisterone should not be used during lactation.

Dosage and administration:

The use of **NORMENS®** is restricted to patients in whom there is no possibility of early pregnancy in the cycle concerned.

Metropathia haemorrhagica (dysfunctional uterine bleeding): 1 tablet 3 times daily for 10 days. Bleeding is arrested usually within 1-3 days. A withdrawal bleeding resembling normal menstruation occurs within 2-4 days after discontinuing treatment.

Prophylaxis against recurrence of dysfunctional bleeding: If there are no signs of resumption of normal ovarian function (no rise in the second half of the cycle of the morning temperature, which should be measured daily) recurrence must be anticipated. Cyclical bleeding can be established with 1 tablet twice daily from the 19th to the 26th day of the cycle.

Premenstrual syndrome (including menstrual mastopathy): Premenstrual symptoms such as headache migraine, breast discomfort, water retention, tachycardia, psychological disturbance may be relieved by the administration of 2-3 tablets daily from the 19th to the 26th day of the cycle. Treatment should be repeated for several cycles. When treatment is stopped, the patient may remain symptom-free for a number of months.

Postponement of menstruation: In case of too frequent menstrual bleeding and in special circumstances (e.g. travel, sports) the Postponement of menstruation is possible. 1 tablet **NORMENS®** three times daily for not longer than 10-14 days, starting 3 days before the expected onset of menstruation. A normal period should occur 2-3 days after the patient has stopped taking tablets.

Endometriosis: Long term treatment is commenced on the 5th day of the cycle with 1 tablet **NORMENS®** twice daily for the first few weeks. In the event of spotting, the dosage is increased to 2 tablets twice daily.

After bleeding has ceased, the initial dose is usually sufficient. Duration of treatment: 4-6 months continuously, or longer if necessary. During treatment, ovulation and menstruation do not occur. After discontinuation of hormone treatment a withdrawal bleeding will occur.

Note: If menstrual bleeding should fail to follow a course of **NORMENS®**, the possibility of pregnancy must be ruled out before a further course is given.

Contraindications:

Contraindications: Pregnancy. Severe disturbances of liver function. Dubin-Johnson and Rotor syndromes. Previous or existing liver tumours. History during pregnancy of idiopathic jaundice, severe pruritus or herpes gestationis (also known as pemphigoid gestationis). Current thromboembolic processes.

Reasons for immediate discontinuation of the tablets: Occurrence for the first time of migrainous headaches or more frequent occurrence of unusually severe headaches, sudden perceptual disorders (e.g. disturbance of vision or hearing), first signs of thrombophlebitis or thromboembolic symptoms, a feeling of pain and tightness in the chest, pending operations (six weeks beforehand), immobilization (for instance, following accidents), onset of jaundice, hepatitis, general pruritus, significant rise in blood pressure, pregnancy.

Warnings:

If any of the conditions/risk factors mentioned below is present or deteriorates while using **NORMENS®**, an individual risk-benefit analysis should be done before **NORMENS®** is started or continued.

It has been concluded from epidemiological surveys that the use of oral oestrogen/progestogen containing ovulation inhibitors is associated by an increased incidence of thromboembolic diseases. Therefore, one should keep the possibility of an increased thromboembolic risk in mind, particularly where there is a history of thromboembolic diseases.

A patient who develops symptoms suggestive of thromboembolic complications should stop treatment immediately. The need for treatment should be reassessed before continuing therapy.

Women with hypertriglyceridemia, or a family history thereof, may be at increased risk of pancreatitis when using COCs while women with hyperlipidaemia are at increased risk of arterial disease.

In rare cases, benign liver tumours, and even more rarely, malignant liver tumours have been reported in users of hormonal substances such as the one contained in **NORMENS®**. In isolated cases, these tumours have led to life-threatening intra-abdominal haemorrhages. If severe upper abdominal complaints, liver enlargement or signs of intra-abdominal

haemorrhage occur, a liver tumour should be included in the differential diagnosis and, if necessary, the preparation should be withdrawn.

NORMENS[®] can influence carbohydrate metabolism. Parameters of carbohydrate metabolism should be examined carefully in all diabetics before and regularly during treatment.

Chloasma may occasionally occur, especially in women with a history of chloasma gravidarum. Women with a tendency to chloasma should minimise exposure to the sun or ultraviolet radiation when taking **NORMENS[®]**.

Patients who have a history of depression should be carefully observed and the drug discontinued if the depression recurs to a serious degree.

Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Any patient who develops an acute impairment of vision, proptosis, diplopia or migraine headache should be carefully evaluated ophthalmologically to exclude papilloedema or retinal lesions before continuing medication.

Progestogens may cause fluid retention. Special care should be taken when prescribing norethisterone in patients with conditions such as epilepsy, migraine, asthma, cardiac dysfunction and renal dysfunction which might be aggravated by this factor.

If menstrual bleeding should fail to follow a course of **NORMENS[®]**, or if the patient wishes to postpone menstruation in special circumstances, the possibility of pregnancy must be excluded before a further course is given.

Side effects:

Side effects are more common during the first months of initial intake of norethisterone preparations, and subside with duration of treatment. The common side effects are uterine/vaginal bleeding including spotting, hypo-menorrhoea and amenorrhoea (in the treatment of endometriosis), headache, nausea, and oedema. Uncommon side effects include migraine and rarely hypersensitivity reactions, urticaria, rash, visual disturbances and dyspnoea. Dizziness,

aggravation of depression, abdominal pain and cholestasis jaundice are also known to have occurred.

Over dosage:

Symptoms of over dosage include nausea, vomiting, breast enlargement and vaginal bleeding. Gastric lavage may be employed if the over dosage is large and the patient is seen sufficiently early (within 4 hours). There are no special antidotes, and treatment should be symptomatic.

Drug Interaction:

It should be administered with caution when taken together with enzyme inducers (they may accelerate its metabolism).

Pharmaceutical precautions:

Store in cool, dry conditions (below 30°C). Away from light and children.

Package quantities:

Box contains 3x10 and 5x10 tablets in blister strips.

®Trade Mark



Manufactured by:
Renata Limited

Section VII, Milk Vita Road, Mirpur,
Dhaka, Bangladesh.

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